

Aryan Jamshidian

Personal Portfolio

About Me

Hello!

My name is Aryan, and I'm a third-year Engineering Physics student at UBC. My degree gives me a strong background in theoretical physics and math, which I use to understand the details of electromechanical design.

I also dabble in other fields that interest me, such as machine learning and quantum computing. I am very open to broadening my knowledge in new and exciting fields!

In my free time I enjoy rock climbing, hiking, and singing choral music.

Electrical Architecture Redesign

UBC Rover Design Team [Completed]

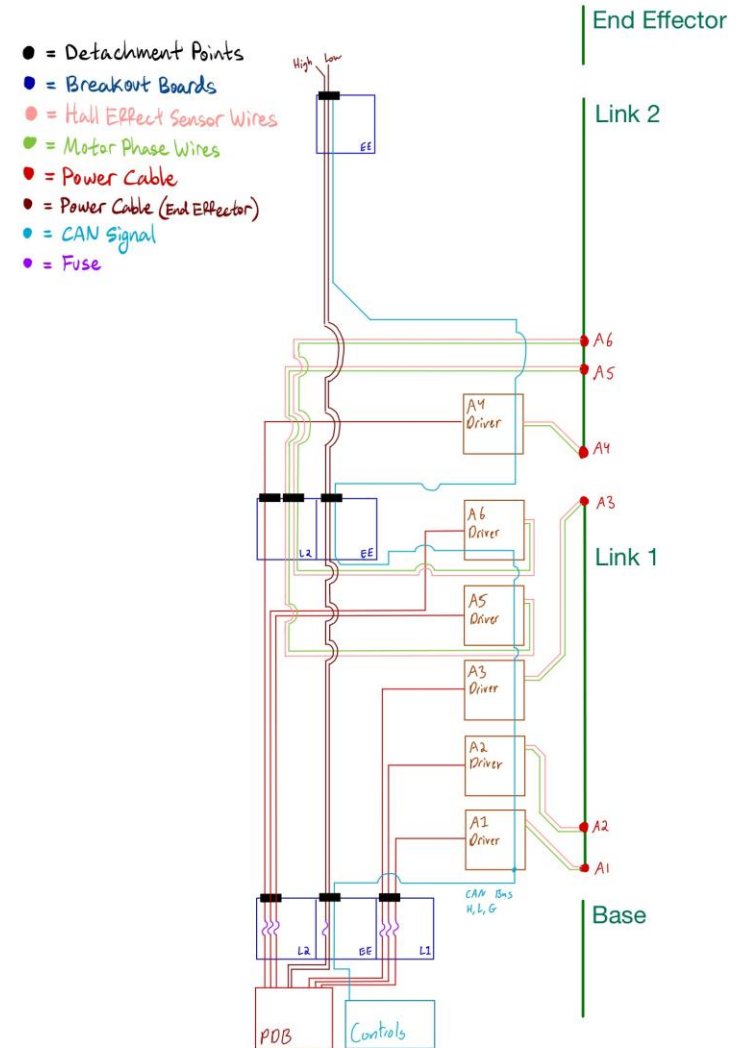
Problem:

- The rover arm has eight degrees of freedom, and extends over a metre in length.
- Running power cables directly from the chassis to every motor driver is messy, difficult to assemble, and almost impossible to test.

Solution:

- Design a new wiring architecture that groups nearby I/O ports together, creates modular disconnection points between every link, and allows for fuses, limit switches, and debugging LEDs.

Wiring Architecture Diagram

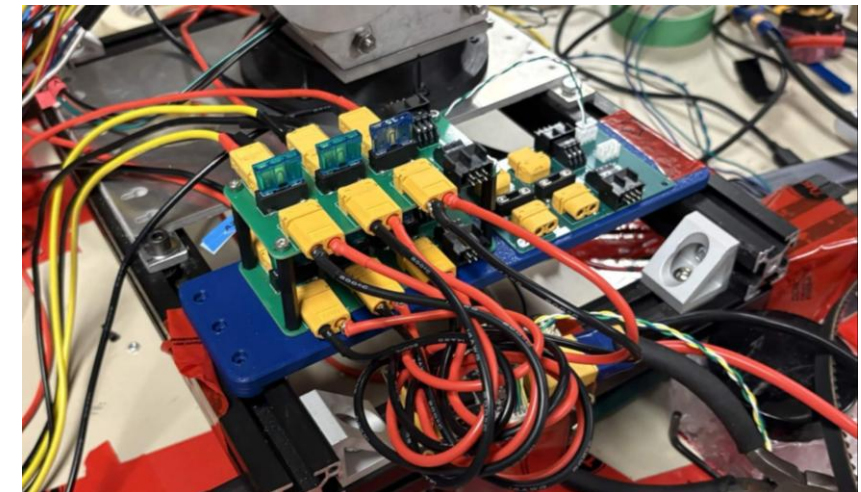
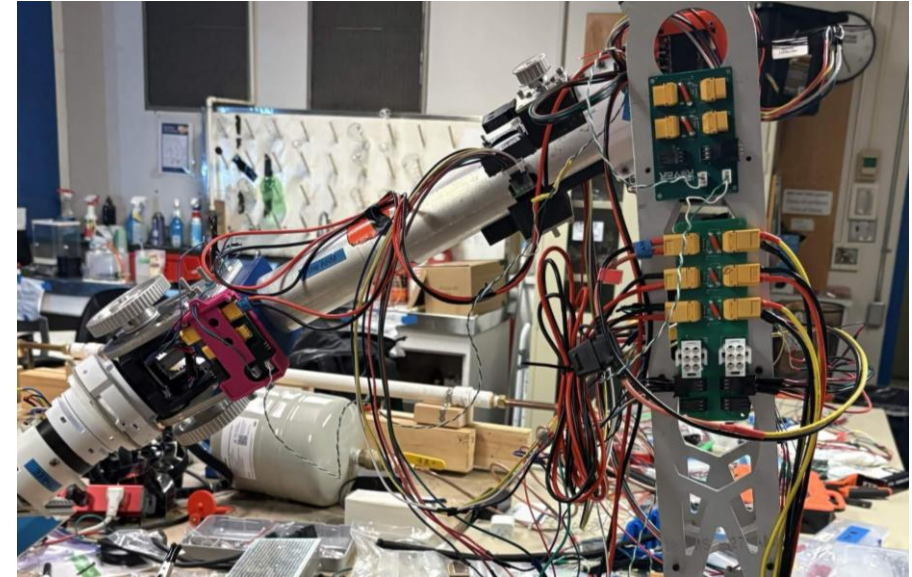


Electrical Architecture Redesign

UBC Rover Design Team [Completed]

Results:

- I designed six custom breakout PCBs in Altium, as well as PCB mounts in SolidWorks for secure and easy access.
- I used impedance matching techniques to minimize signal reflection along the CAN line.
- With the help of my team, we verified PCBs and rewired the arm.
- The project was successful, and is expected to advance the testing schedule by two months while improving quality of life for the arm subteam.



Improving Internal Gear Meshing

UBC Rover Design Team [In-Progress]

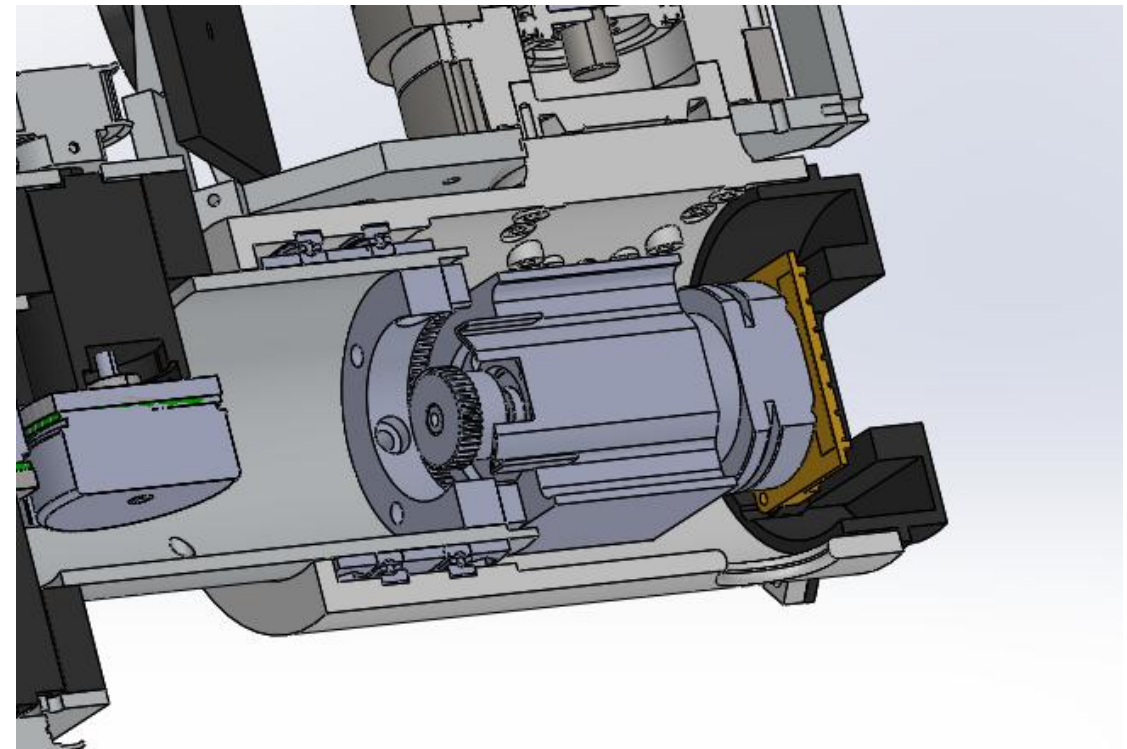
Problem:

- The internal gear that controls the roll joint of the arm is meshing poorly.
- This leads to slippage, and increased wear on the gears.

Solution:

- Isolate and test factors affecting slippage.
- Progress: I eliminated over-toleranced bearings and worn-out spacers as causes.
- Next Steps: I suspect that some linkages may be underconstrained. I will need to do an in-depth analysis of the model.

Roll Joint CAD (Section View)



Mars Habitat Robot Competition

ENPH 253 – Instrument Design [In-Progress]

Goal:

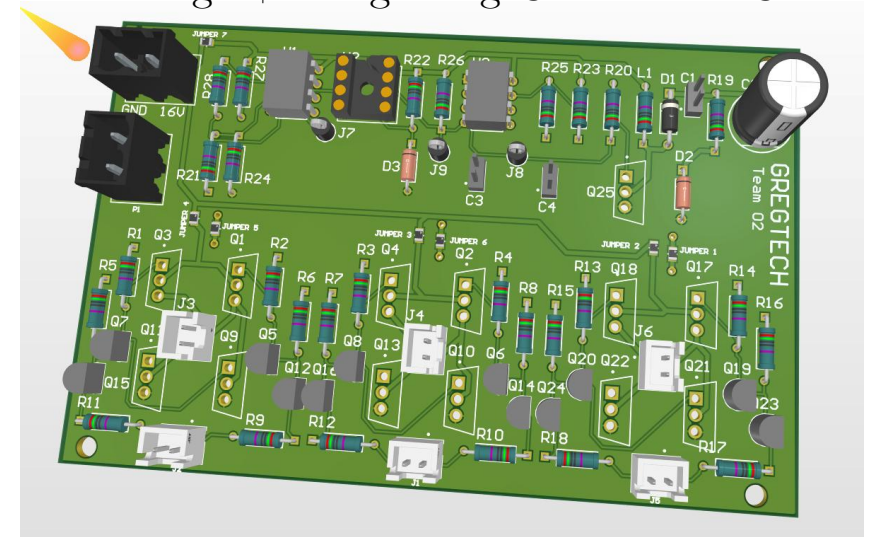
To design a fully autonomous robot that can complete various tasks themed around Mars exploration during a timed competition.

My primary roles in this project are making PCB layouts in Altium, modelling the arm and claw in Onshape, and developing the metal detector circuit.

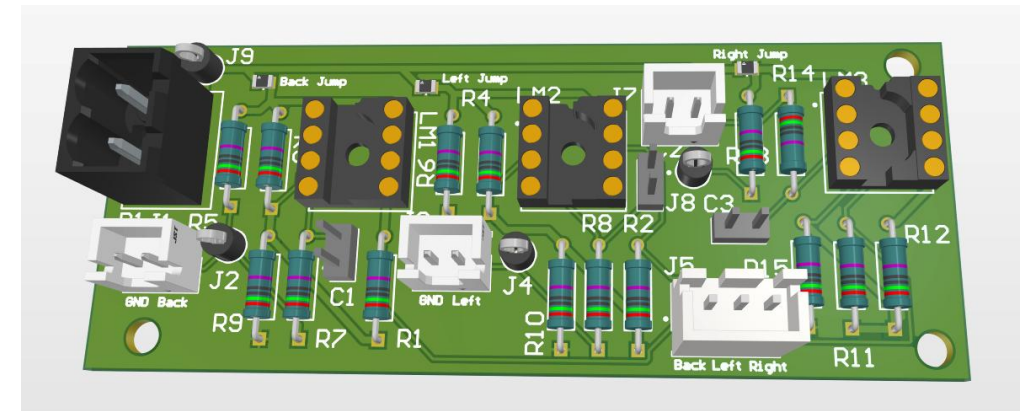
Line Following:

- We are using Kiwi drive to achieve holonomic motion while using less motors than X drive.
- Phototransistor-based PID line following allows us to accurately travel around the field.

H-Bridge + Bang-Bang Controller PCB



Signal Debouncer for Rotary Encoders PCB



Mars Habitat Robot Competition

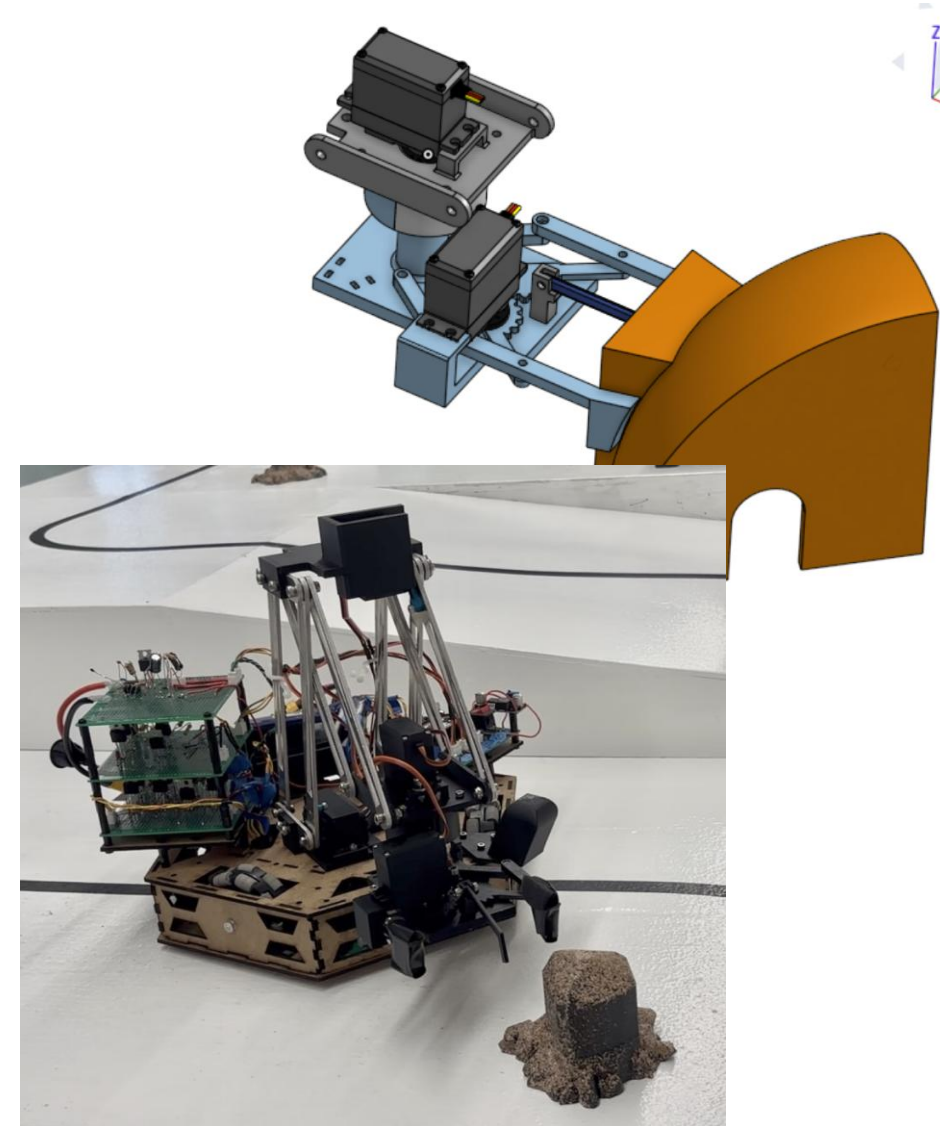
ENPH 253 – Instrument Design [In-Progress]

Arm:

- Uses two 4-bar linkages to keep the claw parallel to the ground while maintaining mobility and reach.
- Two DOFs on the arm allows for full range of motion, rotation of the chassis acts as the third DOF.

Claw:

- Two additional DOFs are yaw and open/close, allowing it to reach around corners and lift objects.
- Pincer geometry is optimized for picking up rocks and aligning with the habitat door.
- Also supports two metal detector coils and a PCB.



Mars Habitat Robot Competition

ENPH 253 – Instrument Design [In-Progress]

Metal Detector:

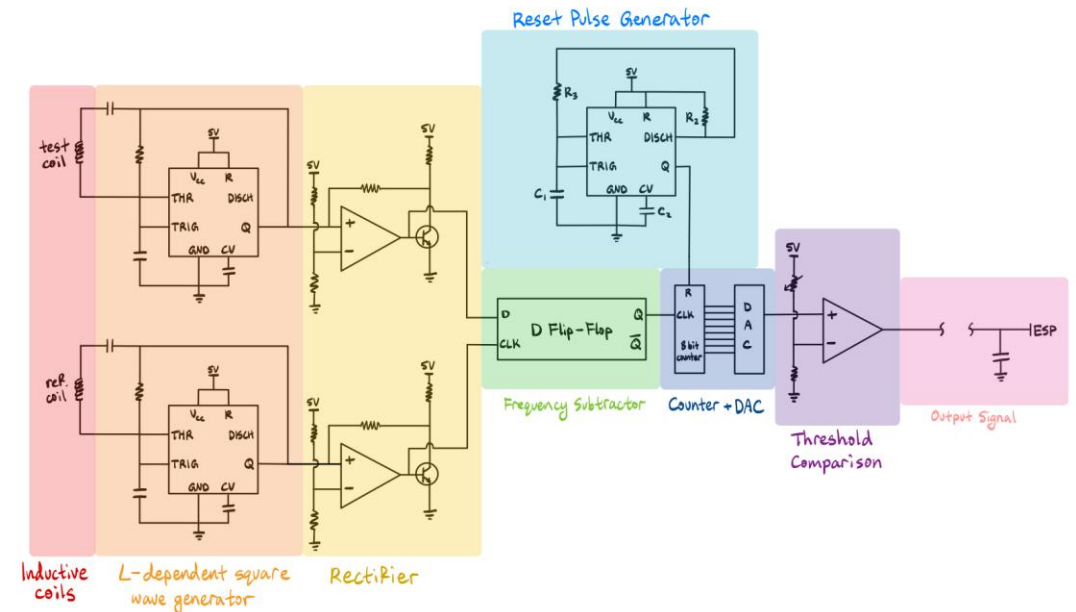
- I designed a custom metal detector circuit to identify aluminum foil hidden in plastic rocks.
- The circuit uses frequency subtraction between two L-dependent 555 timer circuits to detect if one of the coils is exposed to metal.

Computer Vision:

- Our team is testing both OpenCV and YOLO computer vision models to identify small creatures hidden around the field.
- We are running the models on a Raspberry Pi 4, connected to the main ESP32.

Working Principle:

- An inductor (made of coiled wire) will experience a decrease in inductance when exposed to ferro/paramagnetic materials due to eddy currents and polarization.
- Experimental results show that we can detect a $>0.5\%$ change in inductance.



Silent Tokens in Transformer Models

Self-Guided Research Project [Completed]

Problem:

- The current generation of Large Language Models is very good at predicting text, but is still incapable of holding a real conversation.
- The architecture of prompt -> response -> prompt -> response places a hard restriction on the capabilities of conversational models, especially speech models.

Theory:

- LLMs process inputs in the form of tokens. Most of these tokens are words or word stubs, but some of them represent more abstract concepts such as “the start of the user prompt” or “the end of a message”.

Hypothesis:

- Transformer models could be trained to have fluid, uninterrupted conversations by adding a new token, for the concept of “silence”, into the model’s architecture.

Silent Tokens in Transformer Models

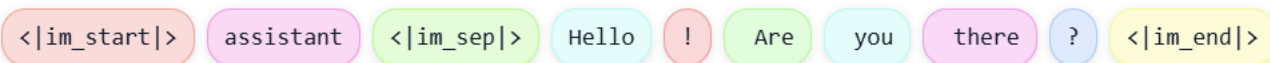
Self-Guided Research Project [Completed]

Procedure:

- I self-taught the fundamentals of machine learning and neural networks, starting from binary classifiers and progressing up to GPT-style self-attention transformers.
- I implemented and modified the GPT architecture, then retrained it on a curated dataset.
- A simplified example of my constructed training set is shown below.

Assistant: Hello! [pauses to wait for response]
Assistant: Are you there? [no reply, so speaks again]

This is a normal interaction with a person that didn't hear/understand you.

A horizontal sequence of colored rounded rectangles representing tokens. From left to right: an orange box with '<|im_start|>', a pink box with 'assistant', a light green box with '<|im_sep|>', a light blue box with 'Hello', a red box with '!', a light green box with 'Are', a light blue box with 'you', a pink box with 'there', a light blue box with '?', and a yellow box with '<|im_end|>'.

This is what gpt-5 would be trained on.

*Since it never sees the [pause], it will learn to say
“Hello! Are you there?” without waiting for a response.

A horizontal sequence of colored rounded rectangles representing tokens. From left to right: an orange box with '<|im_start|>', a pink box with 'assistant', a light green box with '<|im_sep|>', a light blue box with 'Hello', a red box with '!', a grey box with '<|silence|>', a light green box with 'Are', a light blue box with 'you', a pink box with 'there', a light blue box with '?', and a grey box with '<|silence|>'.

This is what my model would be trained on.

Silent Tokens in Transformer Models

Self-Guided Research Project [Completed]

- In theory, the model would learn that after completing a thought, it should output silence tokens (which are relayed to the user as pauses).
- If the user responds, it continues as normal. But if the user doesn't respond, then after a few silence tokens have built up it will learn to start messaging again.
- The number of silence tokens allow it to differentiate between a user responding quickly or slowly.

Results:

- Unfortunately, this experiment did not yield significant results.
- Although the model did sometimes output silence tokens, they did not match the patterns from the training data and mapped to pseudo-random pauses in speech.
- I believe that this idea could be promising with a more suitable model architecture and better training set. I intend to revisit this project in the future.

Contact

Thank you for your interest!

If my experience aligns with your company goals, feel free to contact me with:

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Or learn more about the Engineering Physics program at <https://www.engphys.ubc.ca/>